

Universal Platform Performance Testing - 2024 Q1-Q2

Additional Performance Testing Coverage

(AGS) [Performance Testing Focus Group](#)

(Retirement) [Performance Testing Strategy](#)

Microservice Performance Testing

Tools

1. Jmeter
2. Taurus (Execution)
3. Blazemeter (Reporting)

Coverage

1. Stress Testing
 - a. Determines the robustness of software by testing beyond the limits of normal operation to understand the upper limits of a system's capacity
 - b. Gradually increase number of users capturing results to see changes in response times/errors
2. Load/Endurance Testing
 - a. Understand how a system behaves under an expected load over a period of time
 - b. Based on KPI, run a flow for a number of hours to see variations/spikes in response times/errors or other issues produced over time

Gitlab

Universal Enrollment Scripts Location: <https://gitlab.com/ascensus-gs/products/depot/automation/performance-test-automation/performance/universal-enrollment/jmeter>

Creation & Execution

Additional information and setup can be found in README.md in Gitlab

1. Generate JMX script files through Jmeter (via GUI)
2. Create YAML file for Taurus (to handle reporting to Blazemeter)
3. Local execution can be done through Jmeter (GUI), Jmeter (CLI), or Taurus (for additional reporting)
4. See below for execution for Jenkins execution

Jenkins/Setup

1. Scripts (JMX) and YAML should be placed on loadtestwt1 (login with RSD credentials)
2. Create job in Jenkins (usually handled by devops)
 - a. Example job - http://jenkinsqant1:8080/job/AGSUP_ENDURANCE/
3. Use Jenkins to run YAML file which will execute associated JMX script
 - a. Use YAML to handle users (concurrency) and duration (hold-for)
 - b. NOTE: If Jenkins jobs are stopping early, it is possible there is a docker cleanup script running that is killing jobs going over 2 hours

Notes

1. Initial testing should focus stress testing of individual endpoints (possibly on the same microservice)
2. Subsequent testing can be expanded to be full flows (possibly mimicking front end user activity)
3. Endurance testing can be tested locally but ideally through Jenkins

UI Performance Testing

Lighthouse

Tools

1. Node/Lighthouse
2. Taurus (Execution)
3. Blazemeter (Reporting)

Coverage

1. Performance, quality, and correctness
 - a. Report can include page by page details for performance, accessibility, best practices, etc.

Gitlab

Universal Enrollment Scripts Location: <https://gitlab.com/ascensus-gs/products/depot/automation/performance-test-automation/performance/universal-enrollment/lighthouse>

Creation & Execution

Additional information and setup can be found in README.md in Gitlab

1. Create lighthouse script (recommended through chrome record via devtools)
2. Local execution can be done using node

Notes

1. Execution host may impact results
2. Since universal enrollment is a single page (with DOM changes), lighthouse reporting cannot wait for page loads and requires setup as a flow (similar to selenium)
 - a. Report will be broken up/capture details only as well as written into the flow

Selenium

Tools

1. Selenium
2. Blazemeter browser plugin (recommended for script creation)
3. Taurus (Execution)
4. Blazemeter (Reporting)

Gitlab

Universal Enrollment Scripts Location: <https://gitlab.com/ascensus-gs/products/depot/automation/performance-test-automation/performance/universal-enrollment/taurus>

Creation & Execution

Additional information and setup can be found in [README.md](#) in Gitlab

1. Create selenium script (recommended through blazemeter plugin)
2. Create YAML file for Taurus (to handle reporting to Blazemeter)
3. Local execution will require chromedriver and can be done via Taurus (including Blazemeter reporting)
4. Selenium grid execution can be done with our internal grid: <http://172.25.166.42:4444/wd/hub>

Notes

1. Local execution can be limited due to resources (# of browsers it can launch)
2. Currently selenium grid is running 30 browsers (additional requests can be queued)

Database

Tools

1. Jmeter
2. Taurus (Execution)
3. Blazemeter (Reporting)

Coverage Considerations

1. Frontend user
2. CSR user
3. Feed (incoming/outgoing)
4. APIs (legacy, mobile, universal enrollment)

Gitlab

Universal Enrollment Scripts Location: <https://gitlab.com/ascensus-gs/products/depot/automation/performance-test-automation/performance/database/jmeter>

Creation & Execution

1. Generate JMX script files through Jmeter (via GUI) as necessary
2. Create YAML file for Taurus (to handle reporting to Blazemeter)
3. Local execution can be done through Jmeter (GUI), Jmeter (CLI), or Taurus (for additional reporting)

Notes

1. Testing may be possible through jenkins but it has not be tested/tried